
Research paper

Dr. Ahmed Al-Jarah

Research paper

- A research paper is a formal written document that presents the results of a systematic investigation on a specific topic.
- It is a structured piece of writing where you ask a question, collect information, analyze it, and present your conclusions.
- Research paper must be:
 - Focused Topic:
It studies a specific problem or question
 - Based on Evidence:
Uses data, experiments, or references (books, articles)
 - Analysis and Thinking:
Not just describing, you analyze and explain

Characteristics of High-Quality Research

- Define the problem: Clearly state what you are trying to solve
- Rigorous methods: Use correct and reliable techniques
- Reproducibility: Others should be able to repeat your work and get similar results
- Clarity: Write in a simple and understandable way
- Novelty: Contribute something new or different
- Document code and data: Share and explain your code/data so others can use it

Research paper types

There are several types of research papers, depending on the goal of the study. Most common ones are:

1. Experimental:

- Based on experiments or real data
- Includes results and analysis
- Example: testing a machine learning model

2. Theoretical:

- Focuses on new ideas, models, or theories
- No experiments required
- Example: proposing a new algorithm

3. Survey (Review):

- Summarizes and compares existing research
- Example: review of deep learning methods

• 4. Case Study:

- In-depth analysis of a specific case or application
- Example: AI used in one company or system

• 5. Application:

- Shows how a method is used to solve a real-world problem
- Example: using ML for fraud detection

• 6. Methodology:

- Proposes a new method or improves an existing one
- Example: a faster training technique

Paper structure

Paper structure is the standard way a research paper is organized so readers can easily understand your work.

Common structure:

1. Title: Short and clear description of your work
2. Abstract: Brief summary of the whole paper, Includes: problem, method, and results
3. Introduction : Explains the problem, Why it is important and What your paper will do
4. Related Work: Discusses previous research and Shows how your work is different
5. Methodology: Explains how you solved the problem and Includes models, algorithms, or approach
6. Results: Shows what you found. Uses tables, graphs, or metrics
7. Discussion: Explains what the results mean. Strengths, limitations, insights
8. Conclusion: Summarizes the work and Suggests future improvements
9. References: Lists all sources used

Citation

A citation is a way of giving credit to the original source of information used in your research, mainly covering the following:

- Avoid plagiarism
- Show where your information comes from
- Make your work more credible
- Help readers find the original source

Example:

Machine learning is important [1].

Reference

[1] J. Smith, "Introduction to Machine Learning," 2020.

Citation Styles

- **APA (American Psychological Association)**
 - Example: In-text: (*Smith, 2020*)
 - Reference: Smith, J. (2020). *Title of book*. Publisher.
- **MLA (Modern Language Association)**
 - Example: In-text: (*Smith 25*)
 - Works Cited: Smith, John. *Book Title*. Publisher, 2020.
- **Chicago Style**
 - Footnote: John Smith, *Book Title* (Publisher, 2020), 25.
- **Harvard Style**
 - In-text: (*Smith, 2020*)
 - Reference: Smith, J., 2020. *Book Title*. Publisher.
- **IEEE (Institute of Electrical and Electronics Engineers)**
 - In-text: [1]
 - Reference:[1] J. Smith, *Book Title*, Publisher, 2020.

Publication

There are the two main ways to publish research in Computer Science and Engineering .

Journal Papers

- Longer and more detailed
- Include deep explanations, experiments, and analysis
- Reviewed more strictly (long review process)
- Published in academic journals

Objective: High-quality, complete research

Conference Papers

- Shorter and more focused
- Present new ideas or early results
- Faster review and publication
- Usually presented at a conference

Objective: Share ideas quickly with the community

Writing a paper

- Choose a broad research area that interests you.
- Connect with faculty and students working in that field.
- Attend talks and read research papers.
- Set clear deadlines or milestones and stick to them.

Recommendation

- Combine writing with reading.
- Review the latest research in your topic.
- Keep documenting your literature review.
- Define criteria to compare different studies.
- Organize comparisons in tables.
- Write a well-structured paper in IEEE format and submit it.

Activity 1.1

- Read the research paper carefully and write a short explanation of what you understood.
- The aim of the activity is to learn how to read a research paper, understand the main ideas and express your understanding clearly

Activity 1.2: Research and Presentation on Linear Regression

- Make a search in the IEEE Digital Library to identify a relevant journal paper on linear regression. Select one article and read it thoroughly to understand its main ideas, methodology, and results.
- Based on your analysis, prepare a 10-slide presentation that clearly explains the paper, including its objective, approach, and main result.
- **Objective:**
The purpose of this activity is to develop your ability to locate academic journal papers on a specific topic, critically analyze their content, and effectively communicate the contributions of other researchers through a structured presentation.